

1 Flexible Shaft – Simscape Tutorial



This document provides a simple introduction to using MATLAB and *Simscape* to model the flexible shaft system for ME495. Please consult the official MathWorks training resources for more in-depth coverage of the various topics (e.g., MATLAB, Simulink, and *Simscape*). In a nutshell, *Simscape* is a toolbox within Simulink.

1.1 Open the Model



A zipped file is available from *Canvas* with the *Simscape* model and supporting MATLAB scripts. You'll need to download the file and unzip the folder.

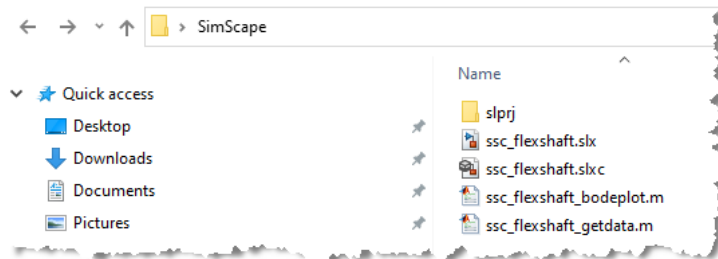
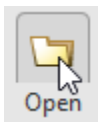


Figure 1-1: File listing of the Simscape folder contents

Step 1. Launch MATLAB.



Step 2. Click the **Open** icon to display the file browser.

Step 3. Navigate to the *Simscape* folder you recently created and select ***ssc_flexshaft.slx*** from the file listing.

The Simulink editor will open with the *Simscape* model as shown in Figure 1-2.

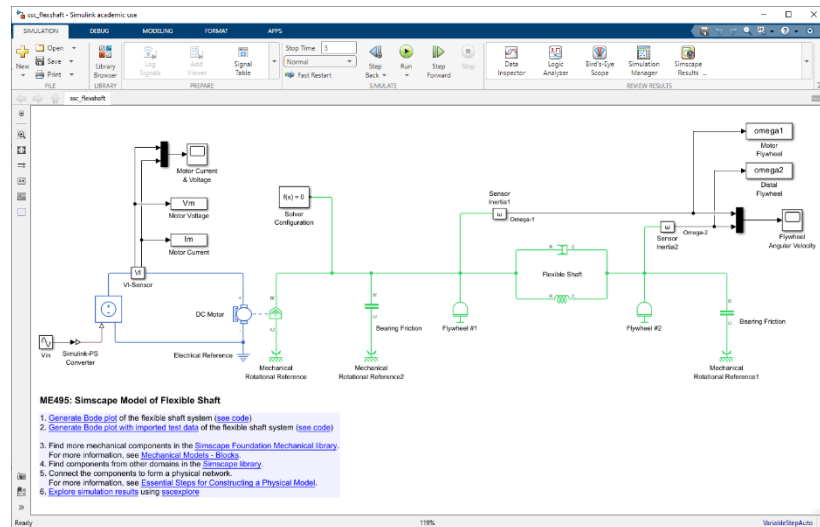


Figure 1-2: Simulink editor with the Simscape model of the Flexible Shaft lab.

1.1 Run the Simulation

In the lower-left corner, click on the [Generate Bode plot](#) hyperlink to automatically run the simulation. The simulation takes approximately one minute to run and generates a Bode Plot based on the current parameter values.

ME495: Simscape Model of Flexible Shaft

1. [Generate Bode plot of the flexible shaft system \(see code\)](#)
2. [Generate Bode plot with imported test data of the flexible shaft system \(see code\)](#)

Figure 1-3: Click the hyperlink to run a simulation

The [Generate Bode plot with imported test data](#) hyperlink opens a file browser to select a data file generated in LabVIEW containing your test results. The experimental results will be plotted with the theoretical model.

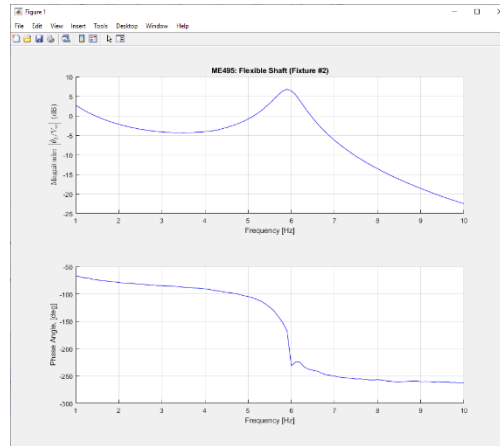


Figure 1-4: Sample Bode plot of the flexible shaft system

1.2 Edit Model Parameters

The model parameters are defined in the MATLAB script entitled *ssc_flexshaft_bodeplot.m*, which can be opened in the MATLAB editor or simply click the [see code](#) hyperlink from the *Simscape* model (see Figure 1-3).

```

ssc_flexshaft_bodeplot.m
[~,~,~,mag,phase] = gettestdata(true);
25 end
26
27
28 %% Construct the model using system parameters
29 K_m = 0.097; % N-m/A or V/(rad/sec)
30 R_m = 1.6; % Ohms
31 L_m = 0.002; % Henry
32 J_m = 4.23e-5; % Kg-m^2
33 D_flywheel = 0.137; % m
34 L_flywheel = 0.0127; % m
35 rho_flywheel = 7755; % kg/m^3
36 D_shaft = 3.18e-3; % m
37 L_shaft = 0.305; % m
38 G_shaft = 7.31e10; % N/m^2
39

```

Figure 1-5: Model parameters defined in MATLAB script
(ssc_flexshaft_bodeplot.m)

1.3 Simscape Basics

1.3.1 Simscape Library



Simscape is a toolbox within the Simulink editor. Click the **Library Browser** icon to display a listing of available toolboxes. As an alternative, click on the tile icon to open the **Simulink Library Browser** in a separate window with the various blocks displayed as tiles.

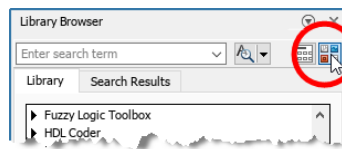


Figure 1-6: Click the tile icon to display block icons

Scroll down the listing and expand the **Simscape** branch to explore the available subject libraries. Although developing a *Simscape* model is beyond the scope of this simple tutorial, you are encouraged to explore the various libraries and blocks.

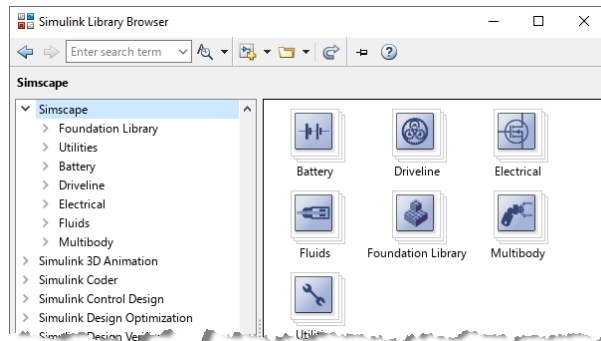


Figure 1-7: Simulink library browser

1.3.2 Flex Shaft Model

The model was created using component blocks from the electrical and mechanical libraries organized under the **Foundation Library**. All the parameters in the model were defined as variables, which are set in the

`ssc_flexshaft_bodeplot` MATLAB script. As an example, double-click on the **Rotational Spring** block to display the parameters.

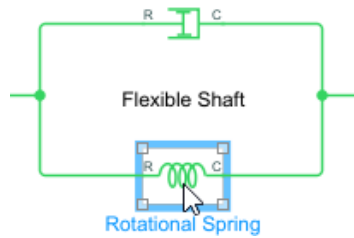


Figure 1-8: Rotational spring block

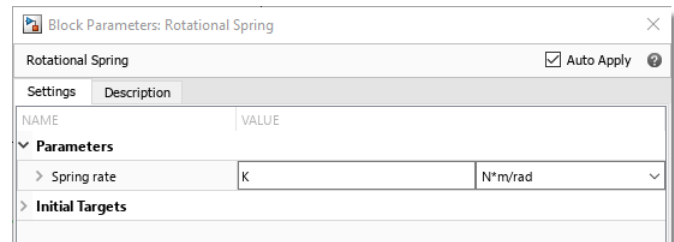


Figure 1-9: Spring rate defined as a parameter (k)

The rotational spring block only has one parameter (i.e. the spring rate, k).

Let's take a look at a more complex block. The **DC Motor** block is defined by electrical and mechanical parameters. Motor armature resistance and inductance are defined with the variables **R_m** and **L_m**, respectively.

Values for these variables are set in the MATLAB script (`ssc_flexshaft_bodeplot`) and passed to the *Simscape* model.

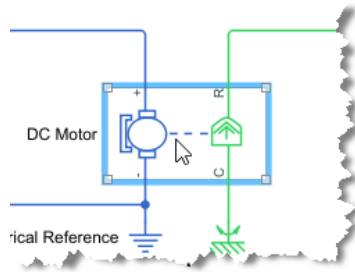


Figure 1-10: DC Motor block

NAME	VALUE
Modeling option	No thermal port
Selected part	<click to select>
Electrical Torque	
Field type	Permanent magnet
Model parameterization	By equivalent circuit parameters
> Armature resistance	R_m Ohm
Armature inductance	L_m H
Define back-emf or torque constant	Specify back-emf constant
> Back-emf constant	K_m V/(rad/s)
Rotor damping parameterization	By damping value
Mechanical	
Rotor inertia	J_m kg*m^2
Rotor damping	B_m N*m*s/rad
> Initial rotor speed	0 rpm
Faults	
Enable armature winding open-circuit f...	Off

Figure 1-11: DC Motor block parameters using variables

The motor voltage and current data are passed between *Simscape*/Simulink and MATLAB using a sensor block. A similar approach is used to capture the rotational velocity of the two flywheels (ω_1 and ω_2).

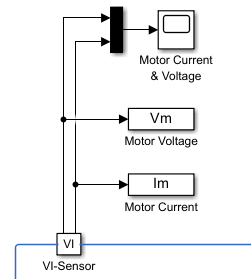


Figure 1-12: Sensor block to pass data to MATLAB

The *Simscape* model provides a more intuitive view of the flexible shaft system than the standard Simulink model, as the model closely resembles the actual physical system. Hopefully, this tutorial has been beneficial in getting you up to speed and enabling you to run a simulation. Feel free to explore and/or modify the model as desired. *Simscape* is an amazing and powerful tool.